王 冲



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研究方向

医学影像处理与分析：加速核磁共振影像（MRI）成像及重建

自然图像与视频处理与分析：图像生成，图像/视频复原，图像增强

教育背景

南洋理工大学， 电子与电气工程，博士 2021/08 - 2025/09

导师：文碧汉，副教授

博士毕业论文：Physics-driven Deep Progressive Reconstruction for Inverse Imaging

爱丁堡大学， 电子与电气工程，硕士 2019/09 - 2020/10

导师：Michael E Davies，教授 (IEEE Fellow)，学位等级：优异（一等荣誉）

硕士毕业论文：Deep Learning for Computational Imaging

电子科技大学， 电子与电气工程，学士 2015/09 - 2019/08

绩点：3.9/4.0

工作经历

博士后，南洋理工大学， 电子与电气工程 2025/12 - 至今

负责项目：Compact Disordered Photonics through Inverse and ML designs

TikTok Pte. Ltd. 字节跳动（新加坡），实习算法工程师 2025/04 - 2025/10

负责项目：基于视觉-语言大模型的图像语义聚类

项目经历

新加坡总理公署国家研究基金会竞争性研究项目: “Harnessing Randomness and Chaos in Complex Disordered Photonic Systems”.（参与） 2024/09 - 至今

发表论文

- **Chong Wang**, Zixuan Fu, Hao Cheng, Siyuan Yang, Alex C. Kot, Bihan Wen, “Taming Noise Initialization for Zero-shot Video Restoration with Image Diffusion Priors”, **IEEE Transactions on Multimedia (TMM) 2026.**
- Fakun Wang*, **Chong Wang***, Yue Zhang, Jianbo Yu, Shi Fang, Wenduo Chen, Bihan Wen, Qi Jie Wang, “An Intelligent Vision Sensor for Hyperspectral Imaging and Computing”, **Nature Communications, 2026. (共同一作)**
- **Chong Wang**, Lanqing Guo, Zixuan Fu, Siyuan Yang, Hao Cheng, Alex C. Kot, Bihan Wen, “Reconciling Stochastic and Deterministic Strategies for Zero-shot Image Restoration using Diffusion Model in Dual”, **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2025.**
- **Chong Wang**, Lanqing Guo, Yufei Wang, Hao Cheng, Yi Yu, Bihan Wen, “Progressive Divide-and-Conquer via Subsampling Decomposition for Accelerated MRI”, **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2024** ***(Highlight, top 2.8%)***.

- **Chong Wang**, Yi Yu, Lanqing Guo, Bihan Wen, "Benchmarking Adversarial Robustness of Image Shadow Removal with Shadow-adaptive Attacks", **IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) 2024** ***(Oral, top 12.7%).***
- **Chong Wang**, Rongkai Zhang, Gabriel Maliakal, Saiprasad Ravishankar, Bihan Wen, "Deep Reinforcement Learning based Unrolling Network for MRI Reconstruction", **IEEE International Symposium on Biomedical Imaging (ISBI) 2023.**
- **Chong Wang**, Rongkai Zhang, Saiprasad Ravishankar, Bihan Wen, "REPnP: Plug-and-Play with Deep Reinforcement Learning Prior for Robust Image Restoration", **IEEE International Conference on Image Processing (ICIP) 2022** ***(Oral, top ~10%)*****.**
- Jianrui Zhang, Zhihao Li, **Chong Wang**, Bihan Wen, "RF4D: Neural Radar Fields for Novel View Synthesis in Outdoor Dynamic Scenes", **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026.**
- Hao Cheng, Siyuan Yang, **Chong Wang**, Joey Tianyi Zhou, Alex C. Kot, Bihan Wen, "STSP: Spatial-temporal Subspace Projection for Video Class-incremental Learning", **European Conference on Computer Vision (ECCV) 2024.**
- Zixuan Fu, Lanqing Guo, **Chong Wang**, Yufei Wang, Zhihao Li, Bihan Wen, "Temporal As a Plugin: Unsupervised Video Denoising with Pre-trained Image Denoisers", **European Conference on Computer Vision (ECCV) 2024.**
- Lanqing Guo, **Chong Wang**, Wenhan Yang, Yufei Wang, Bihan Wen, "Boundary-Aware Divide and Conquer: A Diffusion-based Solution for Unsupervised Shadow Removal", **IEEE/CVF International Conference on Computer Vision (ICCV) 2023.**
- Lanqing Guo, **Chong Wang**, Wenhan Yang, Siyu Huang, Yufei Wang, Hanspeter Pfister, Bihan Wen, "ShadowDiffusion: When Degradation Prior Meets Diffusion Model for Shadow Removal", **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2023.**

在审论文

- **Chong Wang**, Zixuan Fu, Shiqi Huang, Siyuan Yang, Hao Cheng, Bihan Wen, "Persistence Forcing: Exploiting Feature Specialization in Pixel-space Diffusion", **submitted to International Conference on Learning Representations (ICLR).**
- **Chong Wang**, Jiarui Zhang, Siyuan Yang, Hao Cheng, Bihan Wen, "Beyond Camera Views: Lifting Vision Foundation Features into Multimodal LiDAR--Camera Neural Fields", **submitted to AAAI Conference on Artificial Intelligence (AAAI).**
- Siyuan Yang*, **Chong Wang***, Hao Cheng, Jun Liu, Bihan Wen, Alex Kot, Hedvig Kjellstrom, "Image Diffusion Models Are Good 3D Human Motion Generators", **submitted to International Conference on Learning Representations (ICLR). (共同一作)**
- Zixuan Fu, **Chong Wang**, Lanqing Guo, Zhihao Li, Bihan Wen, "Exploiting Diffusion Priors for General Image Restoration with Stochastic Gradient Blocking", **submitted to IEEE Transactions on Multimedia (TMM).**

专业服务

- **期刊审稿人：** IEEE TIP, TMM, TCSVT.
- **会议领域主席:** VCIP.
- **会议审稿人：** CVPR, ICCV, ECCV, NeurIPS, ICLR, AAAI, BMVC, ICASSP, ICME, ISCAS.
- **会议讲座演讲人:** Recent Advances in Machine Learning for Image Reconstruction: Sparse Models to Deep Networks (ISBI 2023).